

Day 8 - Interview Questions

1. What is a string in Python, and how is it defined?

A string is a sequence of characters enclosed in either single (' ') or double (" ") quotes. For example, "Hello, World!" is a string.

2. How can you concatenate (combine) two strings in Python?

You can concatenate two strings using the + operator. For example: "Hello" + " " + "World!" results in "Hello World!".

3. Explain string indexing and slicing.

String indexing allows you to access individual characters within a string using square brackets and a numerical index. String slicing allows you to extract a portion of a string by specifying a range of indices.

4. What is string slicing, and how can you extract a substring from a string in Python?

String slicing allows you to extract a portion of a string by specifying the start and end indices. For example: "Hello"[1:3] returns "el".

5. What is string indexing in Python, and how can you access individual characters in a string?

String indexing allows you to access individual characters in a string. You can use square brackets with an index to access a specific character.

6. How can you traverse a string while skipping every other character?

You can use slicing with a step value of 2 to skip every other character during traversal.

7. Explain how to reverse a string in Python.

You can reverse a string using slicing with a step of -1. For example, reversed_string = string[::-1].

8. What is the difference between '==' and 'is' when comparing strings?

'==' checks if two strings have the same content, while 'is' checks if two string objects are the same (i.e., they reference the same memory location).

9. What is string traversal in Python?

String traversal refers to the process of visiting each character in a string one by one, often to perform some operation on each character.

10. What is the difference between string traversal and string slicing?

String traversal involves iterating through each character in a string, while string slicing involves extracting a portion of a string based on indices.